

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Saturday, 12 September 2026, 9:51 PM**State** Finished**Completed on** Saturday, 12 September 2026, 9:54 PM**Time taken** 2 mins 25 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:

Input: 6

Output: 6

Explanation: There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      long long dp[n + 1];
9
10     for (int i = 0; i <= n; i++)
11         dp[i] = 0;
12
13     dp[0] = 1;
14
15     for (int i = 1; i <= n; i++)
16     {
17         dp[i] = dp[i - 1];
18
19         if (i >= 3)
20             dp[i] += dp[i - 3];
21     }
22
23     printf("%lld", dp[n]);
24
25     return 0;
26 }
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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